

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester VI Examination, 2022-23

Paper No. MTB-611: Dynamical Systems

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Find the evolution operator ϕ_t for $\dot{x} = x^2 - 1$.
- (b) Show that all solutions except the fixed point at origin of the system $\dot{x} = y$, $\dot{y} = -x$ are periodic.
- (c) If P , Q and R are matrices of order n such that $R = PQP^{-1}$, then show that $e^R = Pe^QP^{-1}$.
- (d) For $A = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$, find the exponential matrix e^{At} .
- (e) Is the matrix $\Phi(t) = \begin{pmatrix} 2e^t & -e^{-t} \\ e^t & e^{-t} \end{pmatrix}$ a fundamental matrix for the system $\dot{x} = Ax$? If so, find the matrix A .
- (f) Calculate e^{At} for $A = \begin{pmatrix} 0 & 4 \\ 4 & 0 \end{pmatrix}$.
- (g) Does the non-autonomous equation $\ddot{x} + \left(\frac{2}{t}\right)\dot{x} + x = 0$ ($t > 0$) have bounded solutions?

Question 2.

- (a) State the Lipschitz condition for an initial value problem $\dot{x} = f(t, x)$, $x(t_0) = x_0$. Find the maximum interval of existence for the unique solution of the system $\dot{x} = x^2 + \cos^2 t$ for $t > 0$, $x(0) = 0$. [6]
- (b) Find the evolution operator ϕ_t for the one-dimensional flow $\dot{x} = -x^3$ and show that it forms a dynamical group. [8]

Question 3.

- (a) Solve the system $\dot{x}_1 = x_1 - 2x_2$, $\dot{x}_2 = -3x_1 + 2x_2$ by calculating the fundamental matrix of the system. [7]

- (b) Find the analytical solution of the system:

$$\dot{x} = -y + x(1 - x^2 - y^2), \quad \dot{y} = x + y(1 - x^2 - y^2)$$

by changing it to equivalent polar form.

[7]

Question 4.

- (a) Define an invariant set S for a dynamical system $\dot{x} = f(x)$. Show that the line $y = mx$ is invariant for the system $\dot{x} = Ax$, where A is a constant 2×2 matrix with a simple real eigenvalue λ and corresponding eigenvector $(1, m)^T$, $m \in \mathbb{R}$. [7]
- (b) Determine the stable and unstable subspaces for the linear system $\dot{x} = Ax$, where

$$A = \begin{pmatrix} -2 & -1 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

and show that these subspaces are invariant.

[7]

Question 5.

- (a) If

$$A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

then show that $e^{At} = e^{at} [I \cos(bt) + J \sin(bt)]$, where $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

[7]

- (b) Solve the initial value problem $\dot{x} = Ax$, $x(0) = x_0$, with

$$A = \begin{pmatrix} 0 & -2 & -1 & -1 \\ 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

by using the $A = S + N$ method, where N is a nilpotent matrix.

[7]

Question 6.

- (a) Determine all fixed points of the nonlinear system $\dot{x} = y$, $\dot{y} = x - x^3 - y$ and discuss their stability by linearizing the system. [7]
- (b) Consider the non-autonomous differential equation:

$$\dot{x} = -x + \left(\frac{a}{1+t} \right) x^2, \quad a > 0, \quad t \geq 0$$

Discuss the stability of the critical point $x = 0$ for various values of a .

[7]

Question 7.

- (a) Discuss the existence and stability of all fixed points of the system:

$$\dot{x} = x(1 - x - ay), \quad \dot{y} = b(x - c)y$$

where a , b , and c are non-negative constants. [7]

- (b) Find the solution of the non-homogeneous system:

$$\dot{x} = x + y + t, \quad \dot{y} = -y + 1$$

with the initial conditions $x(0) = 1$, $y(0) = 0$. [7]